

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA1005	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
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Date:	25/07/2026	Duration:	60 minutes

Code of Conduct

I K NISHALINI(192521301) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K NISHALINI

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement

- Explain the problem in your own words.
- Identify the objectives of the proposed system.
- State the scope of the problem.

2. Identify Key Stakeholders and Challenges

- Identify all the stakeholders involved in the system.
- Explain the roles and responsibilities of each stakeholder.
- Discuss the major challenges and issues faced by the stakeholders.

3. Analyze the Current Approach

- Describe the existing system or current process.
- Identify its strengths and limitations.
- Analyze the problems or gaps that need to be addressed.

4. Propose a Suitable Solution Using Software Engineering Principles

- Design a software-based solution for the given problem.
- Describe the major functional modules of the proposed system.
- Identify the functional and non-functional requirements.
- Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

5. Justify the Chosen Methodology/Framework

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. Discuss Expected Outcomes and Limitations

- Describe the expected benefits of the proposed solution.
 - Explain how the solution addresses the identified challenges.
 - Discuss potential risks, implementation challenges, and limitations.
- Suggest possible future enhancements.

AI-ENABLED SMART FOREST FIRE DETECTION AND EMERGENCY RESPONSE SYSTEM

1. Understand the Problem Statement

1.1 Problem Statement (In Your Own Words)

Forest fires are one of the major environmental disasters that destroy forests, wildlife, biodiversity, and human settlements. Traditional fire detection methods mainly depend on watchtowers, satellite images, or reports from local people. These methods often detect fires late, allowing them to spread rapidly. An AI-enabled Smart Forest Fire Detection and Emergency Response System uses Artificial Intelligence (AI), Internet of Things (IoT) sensors, drones, cameras, and weather data to detect fires at an early stage and immediately alert emergency services for faster response.

1.2 Objectives of the Proposed System

- Detect forest fires at an early stage using AI.
- Monitor forest conditions continuously.
- Provide real-time alerts to forest officials and firefighters.
- Predict fire-prone areas using weather and environmental data.
- Reduce damage to forests, wildlife, and nearby communities.
- Improve emergency response time.
- Store historical fire data for future analysis.

1.3 Scope of the Problem

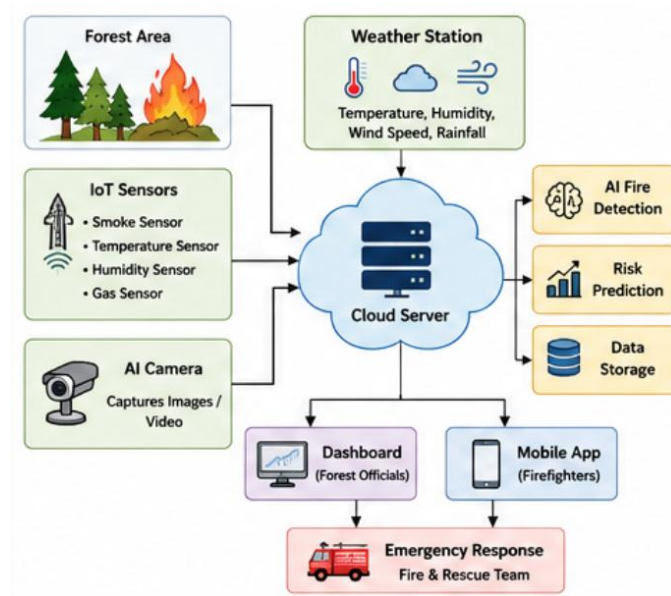
The proposed system covers:

- Continuous forest monitoring
- Smoke and flame detection using AI
- Weather monitoring (temperature, humidity, wind speed)
- GPS-based fire location tracking
- Automatic emergency alerts
- Fire risk prediction
- Dashboard for authorities
- Mobile application for emergency notifications

The system is suitable for:

- National forests
- Wildlife sanctuaries
- Reserved forests
- National parks
- Rural areas near forests

SYSTEM ARCHITECTURE



2. Identify Key Stakeholders and Challenges

2.1 Stakeholders

Stakeholder	Role
Forest Department	Monitor forests and manage fire incidents
Fire & Rescue Department	Extinguish fires
Government Authorities	Policy making and funding
Local Communities	Report incidents and receive alerts
Wildlife Protection Agencies	Protect wildlife
Environmental Organizations	Conservation activities
System Administrators	Maintain the software
AI Engineers	Develop AI detection models
IoT Device Providers	Supply sensors and hardware
Researchers	Analyze fire data

2.2 Roles and Responsibilities

Forest Department

- Monitor forest conditions
- Verify AI alerts
- Coordinate emergency teams

Firefighters

- Respond immediately
- Extinguish fire
- Rescue affected people and wildlife

Government

- Provide budget
- Develop policies
- Ensure implementation

Local Communities

- Report suspicious smoke
- Follow evacuation alerts

AI Engineers

- Train AI models
- Improve prediction accuracy

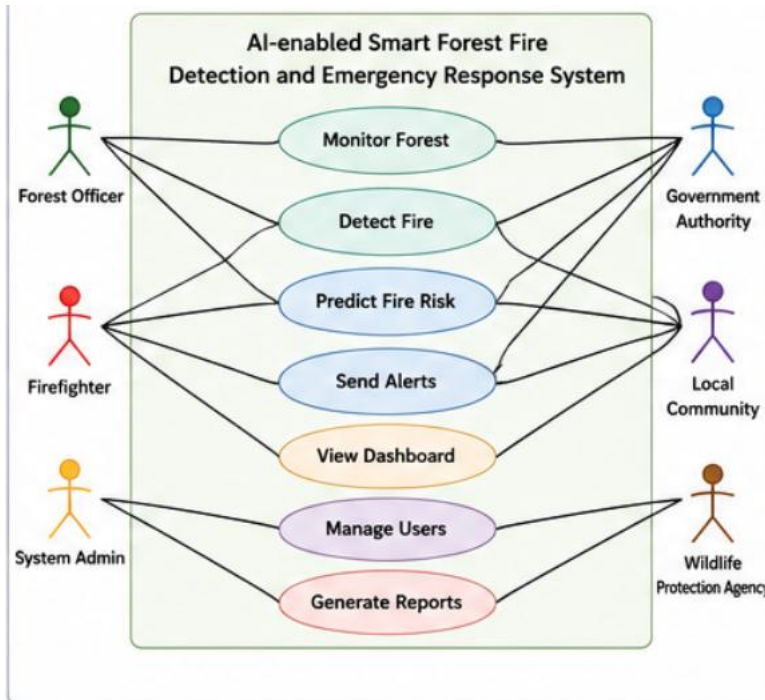
System Administrator

- Maintain servers
- Manage users
- Monitor system performance

2.3 Challenges Faced by Stakeholders

- Delayed fire detection
- Remote forest locations
- Lack of communication network
- False fire alarms
- Limited firefighting resources
- Weather uncertainties
- High operational cost
- Wildlife evacuation difficulties

USE CASE



3. Analyze the Current Approach

3.1 Existing System

Current fire detection methods include:

- Forest watch towers
- Satellite monitoring
- Patrol teams
- Manual reporting
- CCTV cameras
- Public complaints

3.2 Strengths

- Low technology requirement
- Human verification reduces false alarms
- Satellite covers large areas

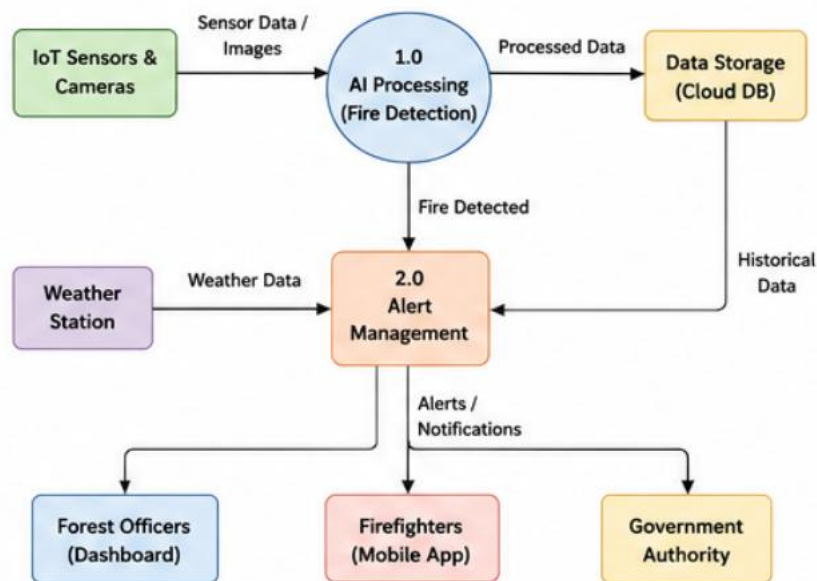
3.3 Limitations

- Slow detection
- Human errors
- Satellite delay
- Weather affects visibility
- Limited night monitoring
- Difficult access to remote forests

3.4 Problems/Gaps

- No real-time monitoring
- Delayed emergency response
- No AI prediction
- Poor coordination
- Lack of centralized monitoring
- No automatic alert generation

DATA FLOW



4. Proposed Solution Using Software Engineering Principles

4.1 Proposed Software Solution

The system combines:

- AI Image Recognition
- IoT Sensors
- Weather Monitoring
- GPS Tracking
- Cloud Computing
- Mobile Notifications
- Emergency Response Management

Whenever smoke or fire is detected:

1. Sensors collect environmental data.
2. Cameras capture images.
3. AI analyzes images.
4. Fire probability is calculated.

5. GPS location is identified.
6. Alert is sent to authorities.
7. Firefighters receive navigation details.
8. Dashboard updates in real time.

4.2 Functional Modules

Module 1: Sensor Monitoring

- Temperature monitoring
- Humidity monitoring
- Smoke detection
- Gas detection

Module 2: AI Fire Detection

- Image processing
- Smoke recognition
- Flame recognition
- Fire classification

Module 3: Risk Prediction

- Weather analysis
- Historical data analysis
- Fire probability prediction

Module 4: Alert Management

- SMS alerts
- Mobile notifications
- Email alerts
- Emergency sirens

Module 5: Emergency Response

- Assign nearest fire station
- Route optimization
- Incident tracking

Module 6: Dashboard

- Fire map
- Live monitoring
- Reports
- Analytics

4.3 Functional Requirements

- User login
- Sensor data collection
- AI fire detection
- GPS tracking
- Real-time dashboard
- Emergency notification
- Fire history management
- Report generation
- User management

4.4 Non-Functional Requirements

- High reliability
- Fast response (<10 seconds)
- High availability (24×7)
- Secure communication
- Easy user interface
- Scalability
- Fault tolerance
- Data backup
- Cloud support

4.5 Software Engineering Principles

Modularity

Separate modules for:

- Detection
- Prediction
- Alerts
- Dashboard
- Reports

Scalability

- Add more sensors easily
- Expand to multiple forests
- Cloud-based deployment

Maintainability

- Easy software updates
- Modular code
- Documentation

Reliability

- Backup servers
- Multiple sensors
- AI verification

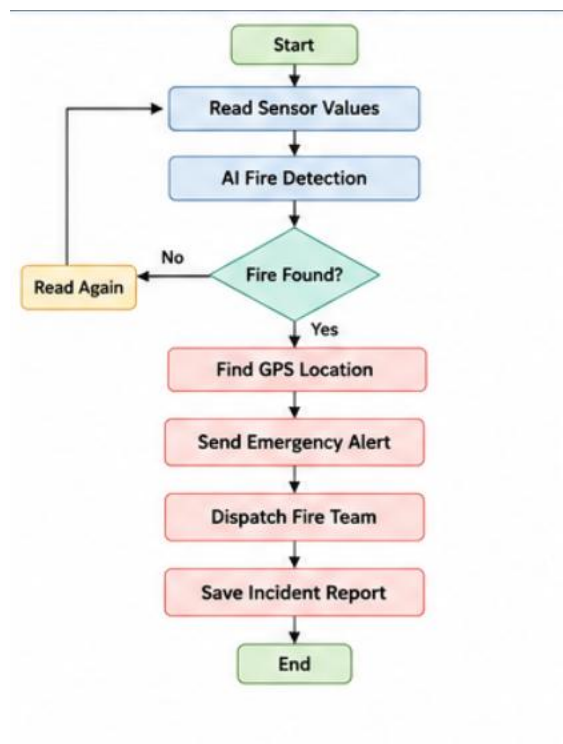
Usability

- Simple dashboard
- Mobile application
- Easy navigation

Security

- User authentication
- Data encryption
- Secure cloud storage
- Role-based access control

FLOW



5. Justify the Chosen Methodology

Selected SDLC Model: Agile (Scrum)

Why Agile?

The project involves:

- AI model improvements
- IoT integration
- User feedback
- Continuous testing
- Feature updates

Agile supports all these effectively.

Advantages of Agile

- Incremental development
- Faster delivery
- Frequent testing
- Customer feedback
- Easy requirement changes
- Reduced project risk

Agile Sprint Example

Sprint	Work
Sprint 1	Requirement gathering
Sprint 2	Sensor integration
Sprint 3	AI model development
Sprint 4	Dashboard development
Sprint 5	Alert system
Sprint 6	Testing
Sprint 7	Deployment

Support for Development & Maintenance

Agile helps by:

- Delivering working software quickly
- Supporting continuous integration
- Improving software quality
- Allowing regular updates
- Encouraging teamwork

6. Expected Outcomes and Limitations

6.1 Expected Benefits

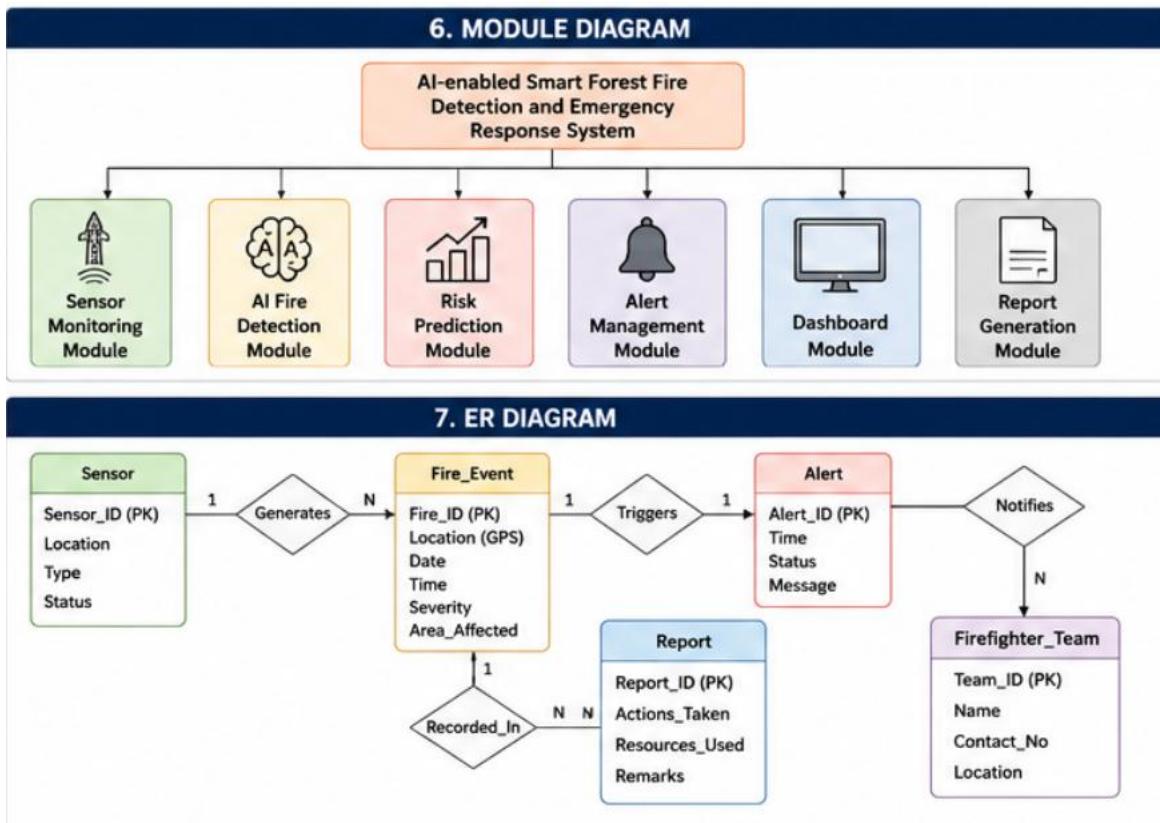
- Early fire detection
- Reduced forest damage
- Faster emergency response
- Better wildlife protection
- Improved public safety
- Lower economic losses
- Better resource management
- Accurate fire prediction
- Real-time monitoring

6.2 How the Solution Addresses Challenges

Challenge	Proposed Solution
Delayed detection	AI real-time monitoring
Remote forests	IoT sensors and drones
Slow communication	Instant alerts
Human errors	AI automation
Poor coordination	Centralized dashboard
Resource shortage	Route optimization

6.4 Future Enhancements

- Drone-based automatic surveillance
- Satellite image integration
- Machine Learning-based fire spread prediction
- Autonomous firefighting robots
- 5G-enabled communication
- Voice-based emergency reporting
- Integration with disaster management systems
- AI-based wildlife evacuation planning
- Renewable energy-powered sensors
- Digital twin simulation for fire management



Conclusion

The AI-enabled Smart Forest Fire Detection and Emergency Response System provides an intelligent, scalable, and reliable solution for protecting forests through early fire detection and rapid emergency response. By integrating AI, IoT, cloud computing, GPS, and real-time communication, the system overcomes the limitations of traditional methods, improves coordination among stakeholders, reduces environmental and economic losses, and supports sustainable forest conservation. Using the Agile (Scrum) methodology ensures continuous improvement, adaptability to changing requirements, and efficient long-term maintenance, making the solution practical for large-scale deployment.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding	/5

Stakeholder & Challenge Analysis	/5
Current System Analysis	/5
Proposed Software Solution & Software Engineering Principles	/5
Methodology/ Framework Justification	/5
TOTAL	/5